

# Higher-Order Metaphysics

## Week Eight: Introduction to model theory for higher order logics

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### 1 Why one might be interested in models

### 2 BBK-models and Muskens models

A **typed collection**  $C$  is a function that maps each type  $\sigma$  to a set  $C^\sigma$ .

When  $C$  and  $D$  are typed collections, a **mapping** from  $C$  to  $D$  is a function  $h$  that maps each type  $\sigma$  to a function  $h^\sigma$  from  $C^\sigma$  to  $D^\sigma$ .

A **variable assignment**  $g$  for a typed collection  $C$  is a mapping to  $C$  from some typed collection of variables.

A variable assignment is **adequate for** a term if it is defined on all variables free in that term.

A **BBK-model** for a signature  $\Sigma$  is a triple  $\mathbf{M} = \langle \mathbf{M}^*, \llbracket \cdot \rrbracket_{\mathbf{M}}, \text{val}_{\mathbf{M}} \rangle$ , where:

- (i)  $\mathbf{M}^*$  is a typed collection of nonempty sets.
- (ii)  $\llbracket \cdot \rrbracket_{\mathbf{M}}$  is a function that maps each type- $\sigma$  term  $A$  of  $\mathcal{L}(\Sigma)$  and variable assignment  $g$  for  $\mathbf{M}^*$  that is adequate for  $A$  to an element  $\llbracket A \rrbracket^g$  of  $\mathbf{M}^\sigma$ , such that the following constraints hold whenever the given assignments are adequate for the given terms:
  - a.  $\llbracket v \rrbracket^g = gv$
  - b. If  $\llbracket A \rrbracket^g = \llbracket C \rrbracket^h$  and  $\llbracket B \rrbracket^g = \llbracket D \rrbracket^h$  then  $\llbracket AB \rrbracket^g = \llbracket CD \rrbracket^h$ .
  - c.  $\llbracket A \rrbracket^g = \llbracket A \rrbracket^h$  when  $g$  and  $h$  agree on all variables free in  $A$ .
  - d.  $\llbracket A \rrbracket^g = \llbracket B \rrbracket^g$  when  $A$  and  $B$  are  $\beta\eta$ -equivalent.<sup>1</sup>
- (iii)  $\text{val}_{\mathbf{M}}$  (the ‘valuation’) is a function from  $\mathbf{M}^t$  to  $\{0, 1\}$ , subject to the following constraints, where ‘ $\mathbf{M}, g \Vdash P$ ’ means ‘ $\text{val}_{\mathbf{M}}[\llbracket P \rrbracket^g] = 1$ ’, and  $g[v \mapsto \mathbf{a}]$  is the function that agrees with  $g$  on variables other than  $v$  and maps  $v$  to  $\mathbf{a}$ :<sup>2</sup>
  - a.  $\mathbf{M}, g \Vdash \neg P$  iff  $\mathbf{M}, g \not\Vdash P$ .
  - b.  $\mathbf{M}, g \Vdash P \wedge Q$  iff  $\mathbf{M}, g \Vdash P$  and  $\mathbf{M}, g \Vdash Q$ .
  - c.  $\mathbf{M}, g \Vdash P \vee Q$  iff  $\mathbf{M}, g \Vdash P$  or  $\mathbf{M}, g \Vdash Q$ .
  - d.  $\mathbf{M}, g \Vdash \forall_\sigma F$  iff  $\mathbf{M}, g[v \mapsto \mathbf{a}] \Vdash Fv$  for every  $\mathbf{a} \in A^\sigma$  ( $v$  not free in  $F$ ).<sup>3</sup>
  - e.  $\mathbf{M}, g \Vdash \exists_\sigma F$  iff  $\mathbf{M}, g[v \mapsto \mathbf{a}] \Vdash Fv$  for some  $\mathbf{a} \in A^\sigma$  ( $v$  not free in  $F$ ).<sup>4</sup>

<sup>1</sup> **BenzmullerBrownKohlhaseHSE** actually only require  $\beta$ , not  $\eta$ .

<sup>2</sup> In condition (iii),  $P$  and  $Q$  are any formulae,  $F$  is any term of type  $\sigma \rightarrow t$ , and  $A$  and  $B$  are any terms of type  $\sigma$ . However, given clause (ii.b), it would make no difference if we restricted the conditions to require  $P, Q, F, A$ , and  $B$  to be variables.

<sup>3</sup> Using a definition below this can be restated as  $\text{ext}_{\mathbf{M}}[\llbracket F \rrbracket^g] = \mathbf{M}^\sigma$ .

<sup>4</sup> Equivalently  $\text{ext}_{\mathbf{M}}[\llbracket F \rrbracket^g] \neq \emptyset$ .

f.  $\mathbf{M}, g \Vdash A = B$  iff  $\llbracket A \rrbracket_{\mathbf{M}}^g = \llbracket B \rrbracket_{\mathbf{M}}^g$ .

A *Muskens model* is defined in the same way as a BBK-model, but dropping clause (ii.d), and adding the following new clause to (iii):

g.  $\mathbf{M}, g \Vdash (\lambda \vec{v}. P) \vec{A}$  iff  $\mathbf{M}, g \Vdash P[\vec{A}/\vec{v}]$

**Theorem.** The class of BBK-models is sound and (strongly) complete for H: for any signature  $\Sigma$ , every theorem of H in  $\mathcal{L}(\Sigma)$  holds in every BBK-model, and every set of sentences consistent in H holds in some BBK-model.<sup>5</sup> Likewise, the class of Muskens models is sound and complete for  $H_0$ .

<sup>5</sup> Indeed every such set holds in some BBK-model in which all the domains are countable.

- the *applicative behaviour*,  $\text{app}_{\mathbf{M}} \mathbf{d}$ , of an element  $\mathbf{d} \in \mathbf{M}^{\sigma \rightarrow \tau}$ , is the function  $\mathbf{a} \mapsto \llbracket Xy \rrbracket^{[X \mapsto \mathbf{d}, y \mapsto \mathbf{a}]}$  from  $\mathbf{M}^{\sigma}$  to  $\mathbf{M}^{\tau}$ .<sup>6</sup>

- The *extension*,  $\text{ext}_{\mathbf{M}} \mathbf{d}$ , of an element  $\mathbf{d} \in \mathbf{M}^{\sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t}$  is the set

$$\{\langle \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \mathbf{M}^{\sigma_1} \times \dots \times \mathbf{M}^{\sigma_n} \mid \mathbf{M}, [X \mapsto \mathbf{d}, y_i \mapsto \mathbf{a}_i] \Vdash Xy_1 \dots y_n\}^7$$

- $\mathbf{M}$  is *functional* iff its applicative behaviour functions are injective for all  $\sigma, \tau$ : that is, for any  $\mathbf{d} \neq \mathbf{d}'$  in  $\mathbf{M}^{\sigma \rightarrow \tau}$ , there is some  $\mathbf{a} \in \mathbf{M}^{\sigma}$  such that  $(\text{app}_{\mathbf{M}} \mathbf{d})\mathbf{a} \neq (\text{app}_{\mathbf{M}} \mathbf{d}')\mathbf{a}$ ; or equivalently, if all instances of Functionality  $(\forall z(Xz = Yz) \rightarrow X = Y)$  hold in it.
- $\mathbf{M}$  is *extensional* iff its extension functions are injective: whenever  $\text{ext}_{\mathbf{M}} \mathbf{a} = \text{ext}_{\mathbf{M}} \mathbf{b}$ ,  $\mathbf{a} = \mathbf{b}$ ; or equivalently, if all instances of Extensionality  $(\forall \vec{z}(X\vec{z} \leftrightarrow Y\vec{z}) \rightarrow X = Y)$  hold in it.
- $\mathbf{M}$  is *Fregean* iff  $\text{val}_{\mathbf{M}}$  is injective; or equivalently, iff  $\mathbf{M}^t$  has exactly two elements; or equivalently, if the Fregean Axiom  $((p \leftrightarrow q) \rightarrow p = q)$  holds in it.
- $\mathbf{M}$  is *functionally full* iff  $\text{app}_{\mathbf{M}}$  is surjective on each domain: every function from  $\mathbf{M}^{\sigma}$  to  $\mathbf{M}^{\tau}$  is the applicative behaviour of some element of  $\mathbf{M}^{\sigma \rightarrow \tau}$ .
- $\mathbf{M}$  is *extensionally full* iff  $\text{ext}_{\mathbf{M}}$  is surjective on each domain: every subset  $X$  of  $\mathbf{M}^{\sigma_1} \times \dots \times \mathbf{M}^{\sigma_n}$  is the extension of some element of  $\mathbf{M}^{\sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t}$ .

*Fact:* so long as  $e$  isn't a terminal type, extensionality is equivalent to the combination of functionality and Fregeanness.

*Fact:* functional fullness entails extensional fullness.

*Fact:* so long as  $e$  isn't a terminal type, the combination of extensional fullness and extensionality entails functional fullness.

<sup>6</sup> By clauses (ii.b) and (ii.c), it doesn't matter which variables we pick to be  $X$  and  $y$ . **BenzmullerBrownKohlhaseHSE** treat the application map as a separate ingredient in the definition of "model", but since it can be recovered from  $\llbracket \cdot \rrbracket$  we omit it.

### 3 Full possible-worlds models

The below all relies on using the type system where  $e$  isn't a terminal type.

For two nonempty sets  $X, C$ , and  $W_0 \in C$ , the *full intensional* structure on  $X, C, W_0$  is the typed collection  $\mathbf{M}$  where  $\mathbf{M}^e = X$  and for any  $\vec{\sigma} = \sigma_1, \dots, \sigma_n$  ( $n \geq 0$ ),  $\mathbf{M}_X^{\vec{\sigma} \rightarrow t}$  is the powerset of  $C \times \mathbf{M}_X^{\sigma_1} \times \dots \times \mathbf{M}_X^{\sigma_n}$ .

To turn this into a model, we need one extra thing: an *interpretation function*  $\mathcal{I}$  which maps each nonlogical constant of type  $\sigma$  to an element of  $\mathbf{M}_X^\sigma$ . We can then define  $\llbracket \cdot \rrbracket$  as follows:

$$\begin{aligned}
\llbracket v \rrbracket^g &= gv & \llbracket c \rrbracket &= \mathcal{I}c \\
\llbracket \wedge \rrbracket &= \{ \langle w, \mathbf{p}, \mathbf{q} \rangle \mid w \in \mathbf{p} \cap \mathbf{q} \} & \llbracket \vee \rrbracket &= \{ \langle w, \mathbf{p}, \mathbf{q} \rangle \mid w \in \mathbf{p} \cup \mathbf{q} \} \\
\llbracket \neg \rrbracket &= \{ \langle w, \mathbf{p} \rangle \mid w \notin \mathbf{p} \} & \llbracket =_\sigma \rrbracket &= \{ \langle w, \mathbf{a}, \mathbf{a} \rangle \mid \mathbf{a} \in \mathbf{M}_X^\sigma \} \\
\llbracket \forall_\sigma \rrbracket &= \{ \langle w, \mathbf{f} \rangle \mid \langle w, \mathbf{a} \rangle \in \mathbf{f} \text{ for all } \mathbf{a} \in \mathbf{M}_X^\sigma \} & \llbracket \exists_\sigma \rrbracket &= \{ \langle w, \mathbf{f} \rangle \mid \langle w, \mathbf{a} \rangle \in \mathbf{f} \text{ for some } \mathbf{a} \in \mathbf{M}_X^\sigma \} \\
\llbracket A^{\sigma \rightarrow t} B^\sigma \rrbracket^g &= \{ \langle w, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \mid \langle w, \llbracket B \rrbracket^g, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket^g \} & \llbracket \lambda v^\sigma . A^\tau \rrbracket^g &= \{ \langle w, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \mid \langle w, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket^{g[v \mapsto \mathbf{a}_0]} \}
\end{aligned}$$

Finally we set  $\text{val } \mathbf{p} = 1$  iff  $W_0 \in \mathbf{p}$ .

We need to confirm that this is a model. This is mostly trivial, except for the part that requires that  $\beta\eta$ -terms have the same denotation. For this, we need to first prove (by an induction on the construction of terms) the

*Substitution lemma:*  $\llbracket A[B/v] \rrbracket^g = \llbracket A \rrbracket^{g[v \mapsto \llbracket B \rrbracket^g]}$ .

*Replacement lemma:* if  $\llbracket A \rrbracket^g = \llbracket B \rrbracket^g$  for all  $g$ , then  $\llbracket C[A/v] \rrbracket^g = \llbracket C[B/v] \rrbracket^g$  for all  $C, v, g$ .

Then we argue that

$$\begin{aligned}
\llbracket (\lambda v.A)B \rrbracket^g &= \{ \langle w, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \mid \langle w, \llbracket B \rrbracket^g, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket \lambda v.A \rrbracket^g \} \\
&= \{ \langle w, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \mid \langle w, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket^{g[v \mapsto \llbracket B \rrbracket^g]} \} \\
&= \llbracket A \rrbracket^{g[v \mapsto \llbracket B \rrbracket^g]} \\
&= \llbracket A[B/v] \rrbracket^g
\end{aligned}$$

And when  $v$  isn't free in  $A$ ,

$$\begin{aligned}
\llbracket \lambda v.Av \rrbracket^g &= \{ \langle w, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \mid \langle w, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket Av \rrbracket^{g[v \mapsto \mathbf{a}_0]} \} \\
&= \{ \langle w, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \mid \langle w, \llbracket v \rrbracket^{g[v \mapsto \mathbf{a}_0]}, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket^{g[v \mapsto \mathbf{a}_0]} \} \\
&= \{ \langle w, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \mid \langle w, \mathbf{a}_0, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket^g \} \\
&= \llbracket A \rrbracket^g
\end{aligned}$$

We can also immediately see that it's a model of C5 + Atomicity.<sup>8</sup> Every instance of Logical Equivalence holds, since if  $H \vdash P \leftrightarrow Q$ ,  $P \leftrightarrow Q$  is true on every assignment in every world, so  $\llbracket \lambda \vec{v}.P \rrbracket^g = \llbracket \lambda \vec{v}.Q \rrbracket^g$  for every assignment, so  $\llbracket \lambda \vec{v}.P = \lambda \vec{v}.Q \rrbracket^g$  includes every world.

<sup>8</sup> Recall that C5 is Classicism +  $\Box ND$

Similarly, whenever  $\llbracket x \rrbracket^g \neq \llbracket y \rrbracket^g$ , there is no world  $w$  such that  $\langle w, \llbracket x \rrbracket^g, \llbracket y \rrbracket^g \rangle \in \llbracket =_\sigma \rrbracket$ , so  $\llbracket x = y \rrbracket^g = \emptyset$ , so  $\llbracket x \neq y \rrbracket^g = W$ , so  $\llbracket (x \neq y) = \top \rrbracket^g = W$ . So for any  $g$  at all,  $\llbracket x \neq y \rightarrow \Box(x \neq y) \rrbracket^g = W$ ; so also for any  $g$ ,  $\llbracket \forall x \forall y. x \neq y \rightarrow \Box(x \neq y) \rrbracket^g = W$  and  $\llbracket \Box \forall x \forall y. x \neq y \rightarrow \Box(x \neq y) \rrbracket^g = W$ .

And for Atomicity, note that  $\text{Atom } q$  is true at  $w$  on  $g$  iff  $gq$  is a singleton, while  $\Diamond p$  is true at  $w$  on  $g$  iff  $gp$  is nonempty. So if  $\Diamond p$  is true at  $w$  on  $g$ , there is an assignment  $g^+$  (that maps  $q$  to a singleton subset of  $gp$ ) on which  $\text{Atom } q \wedge (q \leq p)$  is true.

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## 5 Full action models

We are interested in finding a more flexible notion of model on which we can have models of theories inconsistent with C5+Atomicity. There are two separate axes of generalization. First, we need to allow *non-fullness*, where not every set of appropriate tuples has to be in the domain of a relational type. And second, we need some tweak to the possible-worlds thing that allows for failures of ND and BF. Let's discuss them in reverse order.

The key to the second axis of generalization is the handy mathematical concept of a *category*. For present purposes, a category  $C$  is a tuple comprising: (i) a set called the set of *objects*; (ii) another set called the set of *arrows*; (iii) a pair of functions  $\text{src}$  and  $\text{trg}$  that assign each arrow a "source" and "target" object; (iv) a partial function  $\circ$  that maps any pair of arrows  $h$  and  $i$ , where the source of  $h$  is the target of  $i$ , into a new arrow  $h \circ i$  which shares  $i$ 's source and  $h$ 's target; and (v) a function  $1$  that maps each object  $A$  into an arrow  $1_A$  whose source and target are  $A$ , such that the following laws hold whenever defined:

$$h \circ (i \circ j) = (h \circ i) \circ j$$

$$h \circ 1_A = h$$

$$1_B \circ h = h$$

Some examples of categories:

- A category with one object  $W_0$  and one arrow  $1_{W_0}$ .
- A category with one object  $W_0$  and two arrows  $1_{W_0}$  and  $k$ , such that  $k \circ k = 1_{W_0}$ .
- A category with one object  $W_0$  and two arrows  $1_{W_0}$  and  $k$ , such that  $k \circ k = k$ .

- A category with two objects  $W_0$  and  $W_1$  and three arrows,  $1_{W_0}$ ,  $1_{W_1}$ , and  $k : W_0 \rightarrow W_1$ .
- For any preordered set, a category whose objects the elements of the set and whose arrows are the ordered pairs in the partial order  $\leq$ , with  $1_A = \langle A, A \rangle$  and  $\langle A, B \rangle \circ \langle B, C \rangle = \langle A, C \rangle$ .

An *action*  $-^*$  on a category  $\mathcal{C}$  is a function that assigns to every object  $A$  of  $\mathcal{C}$  a set, and to every arrow  $h : A \rightarrow B$  of  $\mathcal{C}$  a function  $h^*$  from  $A^*$  to  $B^*$ , in such a way that  $(h \circ i)^* = h^* \circ i^*$  and  $1_A^*$  is the identity function on  $A^*$ .

The *powerset action*  $-^{\mathcal{P}}$  on a category  $\mathcal{C}$  is the action such that

- For each object  $A$ ,  $A^{\mathcal{P}}$  is the set of all sets of arrows with source  $A$ .
- For each arrow  $h : A \rightarrow B$  and  $\mathbf{p} \in A^{\mathcal{P}}$ ,  $h^{\mathcal{P}} \mathbf{p} = \{i \mid i \circ h \in \mathbf{p}\}$ .

More generally, when  $-^{*1}, \dots, -^{*n}$  are any actions on  $\mathcal{C}$ , the *intension action*  $-^{[*1, \dots, *n]}$  is the action such that

- For each object  $A$ ,  $A^{[*1, \dots, *n]}$  is the set of  $n$ -tuples  $\langle h, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle$  where for some object  $B$ ,  $h : A \rightarrow B$  and each  $\mathbf{a}_i$  is a member of  $B^{*i}$ .
- For each arrow  $h : A \rightarrow B$  and  $\mathbf{r} \in A^{[*1, \dots, *n]}$ ,  $h^{[*1, \dots, *n]} \mathbf{r} = \{\langle i, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \mid \langle i \circ h, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \mathbf{r}\}$ .

Given a base category  $\mathcal{C}$  and an action  $-^*$  on  $\mathcal{C}$ , we can define a family of actions  $-^\sigma$  for every type  $\sigma$ —the *full intensional structure* over  $\mathcal{C}$  and  $-^*$ —as follows:

$$\begin{aligned} -^c &= -^* \\ -^{\sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t} &= -^{[\sigma_1, \dots, \sigma_n]} \end{aligned}$$

To turn this into a model, we need two more things: a distinguished object  $W_0$  of  $\mathcal{C}$ , for which we'll set the type- $\sigma$  domain of the model to be  $W_0^\sigma$ ; and an interpretation function  $\mathcal{I}$  that maps each nonlogical constant  $c$  of type  $\sigma$  to an element of  $W_0^\sigma$ .

We define the interpretation function of our model with the help of a more complex function  $\llbracket \cdot \rrbracket$ : that takes a type- $\sigma$  term  $A$ , an arrow  $h$  with source  $W_0$  and any target object  $V$  and an assignment function  $g$  for  $V$  and yields an element of  $V^\sigma$ . This is defined as follows.

$$\begin{aligned} \llbracket v \rrbracket_h^\sigma &= gv & \llbracket c^\sigma \rrbracket_h &= h^\sigma \mathcal{I} c \\ \llbracket \wedge \rrbracket_h &= \{\langle i, \mathbf{p}, \mathbf{q} \rangle \mid 1_{\text{trg } i} \in \mathbf{p} \cap \mathbf{q}\} & \llbracket \vee \rrbracket_h &= \{\langle i, \mathbf{p}, \mathbf{q} \rangle \mid 1_{\text{trg } i} \in \mathbf{p} \cup \mathbf{q}\} \\ \llbracket \neg \rrbracket_h &= \{\langle i, \mathbf{p} \rangle \mid 1_{\text{trg } i} \notin \mathbf{p}\} & \llbracket =_\sigma \rrbracket_h &= \{\langle i, \mathbf{a}, \mathbf{a} \rangle \mid \mathbf{a} \in (\text{trg } i)^\sigma\} \\ \llbracket \forall_\sigma \rrbracket_h &= \{\langle i, \mathbf{f} \rangle \mid \langle 1_{\text{trg } i}, \mathbf{a} \rangle \in \mathbf{f} \text{ for all } \mathbf{a} \in (\text{trg } i)^\sigma\} & \llbracket \exists_\sigma \rrbracket_h &= \{\langle i, \mathbf{f} \rangle \mid \langle 1_{\text{trg } i}, \mathbf{a} \rangle \in \mathbf{f} \text{ for some } \mathbf{a} \in (\text{trg } i)^\sigma\} \\ \llbracket A^{\sigma \rightarrow t} B^\sigma \rrbracket_h^\sigma &= \{\langle i, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \mid \langle i, i^\sigma \llbracket B \rrbracket_h^\sigma, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket_h^\sigma\} & \llbracket \lambda v^\sigma. A^\tau \rrbracket_h^\sigma &= \{\langle i, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \mid \langle 1_{\text{trg } i}, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket_{i \circ h}^{(\text{trg } i)^\sigma \rightarrow \tau}\} \end{aligned}$$

We make this be a model by defining  $\llbracket A \rrbracket^g$  as  $\llbracket A \rrbracket_{1_{W_0}}^g$  and setting  $\text{val } \mathbf{p} = 1 \text{ iff } 1_{W_0} \in \mathbf{p}$ .

Say that  $h, g \Vdash P$  iff  $1_{\text{trg } h} \in \llbracket P \rrbracket_h^g$ , and  $g \Vdash P$  iff  $1_{W_0}, g \Vdash P$ . Then we can show that

$$h, g \Vdash \Box_{\top} P \text{ iff } i \circ h, i \circ g \Vdash P \text{ for all } i \text{ composable with } h$$

As before, the only hard part of showing that these things are models is showing that  $\beta\eta$ -conversion holds. For this we need generalizations of the substitution and replacement lemmas, plus the following general fact (proved by induction on complexity):

$$i\llbracket A \rrbracket_h^g = \llbracket A \rrbracket_{i \circ h}^{i \circ g}$$

or to put it differently,

$$\langle j \circ i, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket_h^g \text{ iff } \langle j, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket_{i \circ h}^{i \circ g}$$

We can then argue as follows:

$$\begin{aligned} \llbracket (\lambda v. A) B \rrbracket_h^g &= \{ \langle i, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \mid \langle i, i\llbracket B \rrbracket_h^g, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket \lambda v. A \rrbracket_h^g \} \\ &= \{ \langle i, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \mid \langle 1_{\text{trg } i}, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket_{i \circ h}^{(i \circ g)[v \mapsto i\llbracket B \rrbracket_h^g]} \} \\ &= \{ \langle i, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \mid \langle 1_{\text{trg } i}, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket_{i \circ h}^{i \circ (g[v \mapsto \llbracket B \rrbracket_h^g])} \} \\ &= \{ \langle i, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \mid \langle 1_{\text{trg } i}, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in i\llbracket A \rrbracket_h^{g[v \mapsto \llbracket B \rrbracket_h^g]} \} \\ &= \{ \langle i, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \mid \langle i, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket_h^{g[v \mapsto \llbracket B \rrbracket_h^g]} \} \\ &= \llbracket A \rrbracket_h^{g[v \mapsto \llbracket B \rrbracket_h^g]} \\ &= \llbracket A[B/v] \rrbracket^g \end{aligned}$$

And when  $v$  isn't free in  $A$ ,

$$\begin{aligned} \llbracket \lambda v. Av \rrbracket_h^g &= \{ \langle i, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \mid \langle i, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket Av \rrbracket_{i \circ h}^{(i \circ g)[v \mapsto \mathbf{a}_0]} \} \\ &= \{ \langle i, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \mid \langle 1_{\text{trg } i}, \llbracket v \rrbracket_{i \circ h}^{(i \circ g)[v \mapsto \mathbf{a}_0]}, \mathbf{a}_1, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket_{i \circ h}^{(i \circ g)[v \mapsto \mathbf{a}_0]} \} \\ &= \{ \langle i, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \mid \langle 1_{\text{trg } i}, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket_{i \circ h}^{i \circ g} \} \\ &= \{ \langle i, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \mid \langle 1_{\text{trg } i}, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \in i\llbracket A \rrbracket_h^g \} \\ &= \{ \langle i, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \mid \langle i, \mathbf{a}_0, \dots, \mathbf{a}_n \rangle \in \llbracket A \rrbracket_h^g \} \\ &= \llbracket A \rrbracket^g \end{aligned}$$

In a full action model, ND fails in type  $\sigma$  iff  $h^\sigma$  is non-injective for some arrow  $h$  with source  $W_0$ , and BF fails in type  $\sigma$  iff  $h^\sigma$  is non-surjective for some arrow  $h$  with source  $W_0$ .

(Some examples of this.)

In full action models, Actuality holds because  $\{1_{W_0}\} \in W_0^t$ , and Atomicity holds because every nonempty element of  $W_0^t$  has a singleton subset.

## 6 Non-full models

For each of our notions of “full” model, we can define a corresponding notion of “premodel” on which the domain of the corresponding type only has to be a *subset* of the relevant set.

In an action premodel, we require that for each object  $W$ ,  $W^{\sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t} \subseteq W^{[\sigma_1, \dots, \sigma_n]}$ . We still want each  $-^\sigma$  to be an action, so we require that if  $X \in W^\sigma$  and  $h : W \rightarrow V$ ,  $hX \in V^\sigma$ .<sup>9</sup>

On a premodel, the above definition of  $\llbracket \cdot \rrbracket$  still works but defines a *partial* function:  $\llbracket A \rrbracket_h^g$  might not exist.

A *general action model* is just an action premodel where  $\llbracket A \rrbracket_h^g$  exists for every term  $A$ , assignment  $g$  for  $W_0$ , and arrow  $h$  with source  $W_0$ .

*Theorem:* Classicism is sound and complete for (general) action models.

<sup>9</sup> Similarly, in a possible-worlds premodel, we require that  $M^{\sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t}$  is just some subset of  $M^{\sigma_1} \times \dots \times M^{\sigma_n}$ .

## 7 Coalesced sums